MAT2040
Tutorial 5

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Quick Review

1. Vector (I)
   - Euclidean vector space
   - parametric vector form
2. Vector (II)
   - linear independence
   - connection to linear system solving
   - span
   - spanning set

Question 1

Let $\mathbf{x_1}, \mathbf{x_2}, \mathbf{x_3}$ be linearly independent vectors in $\mathbb{R}^n$ and let

$$\mathbf{y_1} = \mathbf{x_2} - \mathbf{x_1}, \quad \mathbf{y_2} = \mathbf{x_3} - \mathbf{x_2}, \quad \mathbf{y_3} = \mathbf{x_3} - \mathbf{x_1}$$

Are $\mathbf{y_1}, \mathbf{y_2}, \mathbf{y_3}$ linearly independent? Prove your answer.

Solution 1

If $c_1\mathbf{y_1} + c_2\mathbf{y_2} + c_3\mathbf{y_3} = 0$, then

$$c_1(\mathbf{x_2} - \mathbf{x_1}) + c_2(\mathbf{x_3} - \mathbf{x_2}) + c_3(\mathbf{x_3} - \mathbf{x_1}) = 0$$

Rearranging terms

$$(-c_1 - c_3)\mathbf{x_1} + (c_1 - c_2)\mathbf{x_2} + (c_2 + c_3)\mathbf{x_3} = 0$$

Since $\mathbf{x_1}, \mathbf{x_2}, \mathbf{x_3}$ are linearly independent, the coefficients in the equation above must all be 0. Thus we have the linear system

$$\begin{cases} -c_1 - c_3 = 0 \\ c_1 - c_2 = 0 \\ c_2 + c_3 = 0 \end{cases}$$

Solution 1

To solve, we reduce to row echelon form

$$\left[\begin{array}{ccc|c} -1 & 0 & -1 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Since the row echelon form involves free variables, there will be nontrivial solutions. Since there are nontrivial linear combinations of $\mathbf{y_1}, \mathbf{y_2}, \mathbf{y_3}$ that equals to 0, the vectors must be linearly dependent.

Question 2

Find a maximal linearly independent subset of the following set:

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 3 \\ -3 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ -6 \\ 1 \\ -3 \end{bmatrix}, \begin{bmatrix} -4 \\ 3 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -5 \\ 9 \\ -4 \\ -7 \end{bmatrix} \right\}.$$

Solution 2

Assume

$$a_1\begin{bmatrix}1\\0\\-2\\1\end{bmatrix}+a_2\begin{bmatrix}-2\\3\\-3\\4\end{bmatrix}+a_3\begin{bmatrix}3\\-6\\1\\-3\end{bmatrix}+a_4\begin{bmatrix}-4\\3\\1\\2\end{bmatrix}+a_5\begin{bmatrix}-5\\9\\-4\\-7\end{bmatrix}=\begin{bmatrix}0\\0\\0\\0\end{bmatrix},$$

then the augmented matrix is

$$\left[\begin{array}{ccccc|c}1&-2&3&-4&-5&0\\0&3&-6&3&9&0\\-2&-3&1&1&-4&0\\1&4&-3&2&-7&0\end{array}\right]\rightarrow\left[\begin{array}{ccccc|c}1&0&0&-2&0&0\\0&1&0&1&0&0\\0&0&1&0&0&0\\0&0&0&0&1&0\end{array}\right].$$

Solution 2

Here we can see there are only 4 pivots in column 1, 2, 3, and 5, the column 4 is a free variable. So the maximal linearly independent subset for the set S is:

$$S_i = \left\{ \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 3 \\ -3 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ -6 \\ 1 \\ -3 \end{bmatrix}, \begin{bmatrix} -5 \\ 9 \\ -4 \\ -7 \end{bmatrix} \right\}.$$

Question 3

Given

$$\mathbf{x_1} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{x_2} = \begin{bmatrix} 3 \\ 4 \\ 2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ 6 \\ 6 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} -9 \\ -2 \\ 5 \end{bmatrix}.$$

(a) Is $\mathbf{x} \in \text{Span}(\mathbf{x_1}, \mathbf{x_2})$? Why?

(b) Is $\mathbf{y} \in \text{Span}(\mathbf{x_1}, \mathbf{x_2})$? Why?

Solution 3

Consider whether the linear system $a_1\mathbf{x_1} + a_2\mathbf{x_2} = \mathbf{x}$ and $b_1\mathbf{x_1} + b_2\mathbf{x_2} = \mathbf{y}$ have solutions. If not, then the vectors are linearly independent. Otherwise, the vectors are linearly dependent.

(a) Assume $a_1\mathbf{x_1} + a_2\mathbf{x_2} = \mathbf{x}$, write the linear equation into matrix format:

$$\left[\begin{array}{cc|c} -1 & 3 & 2 \\ 2 & 4 & 6 \\ 3 & 2 & 6 \end{array}\right] \rightarrow \left[\begin{array}{cc|c} -1 & 3 & 2 \\ 0 & 10 & 10 \\ 0 & 11 & 12 \end{array}\right] \rightarrow \left[\begin{array}{cc|c} -1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{array}\right].$$

The linear system has no solution, thus $\mathbf{x} \notin \text{Span}(\mathbf{x_1}, \mathbf{x_2})$.

Solution 3

(b) Assume $b_1\mathbf{x_1} + b_2\mathbf{x_2} = \mathbf{y}$, then

$$\left[\begin{array}{cc|c} -1 & 3 & -9 \\ 2 & 4 & -2 \\ 3 & 2 & 5 \end{array}\right] \to \left[\begin{array}{cc|c} -1 & 3 & -9 \\ 0 & 10 & -20 \\ 0 & 11 & -22 \end{array}\right] \to \left[\begin{array}{cc|c} -1 & 3 & -9 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{array}\right].$$

The linear system has a unique solution: $b_1 = 3, b_2 = -2$, thus $\mathbf{y} \in \text{Span}(\mathbf{x_1}, \mathbf{x_2})$.

Question 4

Which of the following are spanning sets for $\mathbb{R}^3$?

(a) $\{(1,1,1)^T, (1,1,0)^T, (1,0,0)^T\}$

(b) $\{(1,2,4)^T, (2,1,3)^T, (4,-1,1)^T\}$

Solution 4

To determine whether a set spans $\mathbb{R}^3$, we must determine whether an arbitrary vector $(a, b, c)^T$ in $\mathbb{R}^3$ can be written as a linear combination of the vectors in the set.

For (a), we must determine whether it is possible to find constants α_1, α_2 and α_3 such that

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \alpha_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

This leads to the system of equations $\begin{cases} \alpha_1 + \alpha_2 + \alpha_3 = a \\ \alpha_1 + \alpha_2 = b \\ \alpha_1 = c \end{cases}$

Since the coefficient matrix of the system is nonsingular, the system has a unique solution. In fact, we find that

$$\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix} = \begin{bmatrix} c \\ b - c \\ a - b \end{bmatrix}$$

Solution 4

Thus,

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = c \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + (b-c) \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + (a-b) \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

so the three vectors span $\mathbb{R}^3$.

For (b), it can be done in the same manner as part (a). If

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} - \alpha_2 \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} + \alpha_3 \begin{bmatrix} 4 \\ -1 \\ 1 \end{bmatrix}$$

then

$$\begin{cases} \alpha_1 + 2\alpha_2 + 4\alpha_3 = a \\ 2\alpha_1 + \alpha_2 - \alpha_3 = b \\ 4\alpha_1 + 3\alpha_2 + \alpha_3 = c \end{cases}$$

Solution 4

In this case, however, the coefficient matrix is singular. Gaussian elimination will yield a system of the form

$$\begin{cases} \alpha_1 + 2\alpha_2 + 4\alpha_3 = a \\ \alpha_2 + 3\alpha_3 = \frac{2a-b}{3} \\ 0 = 2a - 3c + 5b \end{cases}$$

If

$$2a - 3c + 5b \neq 0$$

then the system is inconsistent. Hence, for most choices of a, b and c, it is impossible to express $(a, b, c)^T$ as a linear combination of $(1, 2, 4)^T, (2, 1, 3)^T$ and $(4, -1, 1)^T$. The vectors do not span $\mathbb{R}^3$.